

Radix-4 Booth

signed digits from an overlapping 3-bit window of B

Step 1 — the number

B is an n -bit two's complement value (n even). The top bit counts negative:

$$B = -2^{n-1}B^{n-1} + \sum_{j=0}^{n-2} 2^j B^j$$

Step 2 — the digit

Each digit reads a 3-bit window; neighbouring digits share one bit ($B^{-1} = 0$):

$$B = \sum_{j=0}^{n/2-1} 4^j d_j \quad d_j = -2B^{2j+1} + B^{2j} + B^{2j-1} \in \{-2, -1, 0, 1, 2\}$$

No carry travels between digits: d_j is a local function of its window.

Step 3 — why it is exact

Each B^{2j+1} below the top is counted as $-2 \cdot 4^j$ in d_j and $+4^{j+1}$ in d_{j+1} : net 2^{2j+1} , its weight in Step 1. The top bit is counted once, as $-2 \cdot 4^{n/2-1} = -2^{n-1}$: its negative weight in Step 1. The sign is built into the digit rule.

Step 4 — the product

$$AB = \sum_{j=0}^{n/2-1} 4^j d_j A$$

$d_j A$ takes one of five values:

B^{2j+1}	B^{2j}	B^{2j-1}	d_j	$d_j A$
0	0	0	0	0
0	0	1	1	A
0	1	0	1	A
0	1	1	2	$2A$
1	0	0	-2	$-2A$
1	0	1	-1	$-A$
1	1	0	-1	$-A$
1	1	1	0	0

$n/2$ digits, $n/2$ partial products. Five digit values in radix four: redundant, three control bits, full ± 2 range, no carry.